

Program Question_1

```
Implicit none
Real::x,res
!Interpolated value
Open(Unit=15,file='a3q1b_in.txt')
Open(Unit=17,file='a3q1b_out.txt')
x=8.0
Call ForwardDifference(x,res)
Write(17,*)"Using Forward Difference Formula"
Write(17,*)"Estiamate the river's water level at 8:00 am",x,"is",res
contains
Subroutine ForwardDifference(x,res)
    Implicit none
    Real,Intent(in)::x
    Real,Intent(out)::res
    Real,Dimension(6)::time,level
    Real,Dimension(6,6)::delta
    Real::u,h
    Integer::n,i,j
    !Initialize value
    time=[2.0,6.0,10.0,14.0,18.0,22.0]
    level=[1.2,1.5,1.9,2.3,2.1,1.8]
    n=6
    delta(:,1)=level
    Do j=2,n
        Do i=1,n-j+1
            delta(i,j)=delta(i+1,j-1)-delta(i,j-1)

        End Do
    End Do
    !stepsize
    h=time(2)-time(1)
    u=(x-time(1))/h
    res=level(1)
    Do j=2,n
        res=res+(u*delta(1,j))/factorial(j-1)
        u=u*(u-(j-1))
    End Do
End subroutine
Integer Function factorial(n)
Integer,Intent(in)::n
Integer::i
factorial=1
Do i=1,n
    factorial=factorial*i
End Do
end Function
```

End Program